Paper 007: Prediction, execution, and learning evidence

Theoretical minimum: Paper 007

AuthorityEffect: None. The Apartment 50 preface is fictional staging. Study the guide, attempt these problems before opening the instructor key, and retain your first answer separately from any correction.

Use exact rational arithmetic. Unless a problem changes it, the carrier has vertices a, b, c, ordered directed edges ab, bc, ca, no filled face, claims $h = (1/3, 1/3, 1/3)$, and potential $f = (0, 1/3, 2/3)$. Define

$$(df)_{st} = f_t - f_s, \qquad r = h - df, \qquad C(x) = x_{ab} + x_{bc} + x_{ca}.$$

A potential adjustment changes the representative r; replacing h changes the claims. A supplied execution-status string is data, not an execution.

There are eight instructional levels, three problems per level, four points per problem: 96 points total. The levels order prerequisites; they are not measured learner abilities or age categories. For every answer give the result, a reason, and its scope. No speed requirement is imposed.

Level 0: Declare the lesson

1. **Carrier and convention.** List the vertices, ordered edges, coefficient system, and difference convention. Does drawing the triangle also declare a filled 2-cell? Identify one operation that changes the carrier rather than only the potential.
2. **Exact input.** Compare `Fraction(1, 3)`, `Fraction("1/3")`, and `Fraction(1/3)`. Which two specify the intended rational input directly? Why is an exact fraction made from a float not automatically the intended rational?
3. **Five components.** Outline a lesson containing a declared finite example, worked calculation, learner prediction, changed-input probe, and evidence boundary. Where must the first prediction be recorded? What does a stored notebook answer fail to record?

Level 1: Compute before running

4. **Worked triangle.** Compute all coordinates of df and r, then $C(h)$, $C(df)$, and $C(r)$. State whether all three claims are potential differences.
5. **Constant shift.** Add $7/5$ to each height. Compute the three shifted heights and predict the new df and r without running a cell. Explain the cancellation.
6. **Changed claims.** Replace h by $-h$, keeping f fixed. Compute the residual and its signed cycle sum. Is this residual equal to $-r$? Explain which quantity was changed.

Level 2: Separate an invariant from a repair

7. **General cycle.** For a directed closed walk $v_0, \dots, v_m = v_0$, prove that $\sum_{i=0}^{m-1}(f_{v_{i+1}} - f_{v_i}) = 0$. Deduce the residual-sum identity for arbitrary rational edge claims on that walk. Does testing three potentials prove the identity for every potential?
8. **Completion.** Keep the first two claims at $1/3$ and choose the third to make the cycle sum zero. Give a potential realizing all three completed claims. Is choosing the missing claim the same operation as adjusting a potential on the original claims?
9. **Path versus cycle.** On the path $a \to b \to c$ with claims $(1, 2)$, use $f = (0, 1, 3)$. Add a closing $c \to a$ claim of 1. Compute the closed-cycle residual and sum. Which closing claim would instead make the table exact? Explain why drawing a filled face does not silently zero the original discrepancy.

Level 3: Try to fool the checker

10. **Wrong-sign mutant.** A faulty implementation uses $\widetilde{df}_{st} = f_s - f_t$. Compute $\widetilde{df}$ and $h - \widetilde{df}$. Does checking only $C(\widetilde{df}) = 0$ and $C(h - \widetilde{df}) = 1$ detect it? Give a predicted-coordinate check that does.
11. **Matching LP objectives.** In the toy linear program minimize the non-negative variable x subject to $x \geq 1$, with dual maximize y subject to $0 \leq y \leq 1$, a checker receives $x = 0, y = 0$. Both computed objectives equal zero. Explain why this is not an optimality certificate, and give the missing feasibility checks and a valid optimal pair.
12. **Status-string grade.** The legacy function counts code cells whose source contains the substring `assert` and treats supplied status `"ok"` as a pass. Evaluate it on:
    - `# assert example` with status `"ok"`;
    - `print("assert")` with status `"ok"`;
    - `assert False` with status `"error"`. Give the reported fraction and explain two reasons it is not a measure of learning or fresh execution.

Level 4: Reproduce the right bytes

13. **Stored outputs versus execution.** A test reads notebook JSON and finds six code cells, execution counts 1 to 6, and the expected output strings. Has it executed the notebook? Describe a fresh replay that compares outputs without overwriting the deposit, and one claim even a successful replay does not support.
14. **Historical source move.** A cited file at commit A is missing locally, while a file with the same name exists at current commit B with different bytes. State a permissible replay strategy, an impermissible substitution, and the information to record if A's bytes cannot be obtained.
15. **Historical 20/20.** The route receipt reports 20 passes dated 2026-09-25; its manifest paths refer to the former notebook tree, and one lesson calls a retired norm API. What can the saved receipt support? What must be maintained and rerun before asserting a fresh full-route pass? Does passing the standalone fraction notebook resolve those route blockers?

Level 5: Read the formal statement

16. **List bound.** The Lean bound assumes $n \in \mathbb{N}$, an integer list h, `h.length n`, and $|x| \leq n - 1$ for every entry; subtraction in `Nat` is truncated at zero. Compute the list mass and ceiling for $n = 3, h = (2, -2, 1)$. Show a permitted list that saturates the ceiling. State the consequence at $n = 0$ and $n = 1$.
17. **Missing hypotheses.** For $n = 3$, give one list violating the per-entry cap and another violating only the length bound, each with mass greater than $n(n-1)$. Explain why neither refutes the conditional theorem and why it cannot be read as a human cognitive limit.
18. **Declaration census.** A saved named report lists 21 declarations; the saved full package-local report lists 64, including generated/private names. Which count answers each scope? Does a whitelist containing `Classical.choice` show that it was used? Does a saved report establish a fresh kernel run on this host? Explain what unary nesting preservation does and does not bound.

Level 6: Assess a person

19. **Threshold fixture.** The legacy test accepts supplied pass rate 92/100, transfer index 88/100, and efficiency 85/100 under chosen thresholds $9/10, 4/5, 4/5$. Calculate which comparisons pass. What participant evidence and definitions are absent? What would be wrong with calling the result a measured 90% learner pass rate?
20. **Transfer comparison.** Propose a comparison of full worked examples with progressively incomplete examples. Specify baseline assessment, assignment/comparison conditions, independently scored novel tasks, and a delayed assessment. Identify one way that answer-key exposure can

contaminate the comparison.

21. **First-response record.** Design a minimal learner record distinguishing first prediction, explanation, output exposure, correction, and later transfer response. How should a missing delayed response be represented? Why should a correction after seeing the answer not replace the first score?

Level 7: Defend the publication's claim

22. **Lesson redesign.** A notebook sets `expected = 1`, assigns `measured = expected`, and asserts equality under the title "all students transfer cycle reasoning." Replace it with a five-component triangle lesson, including an independently derived reference and a control that can detect a specific error. Write its strongest justified evidence sentence.
23. **Receipt audit.** A release verifier reports that file hashes and the two-way file census match. A fraction notebook replay passes. Lean is unavailable. Write a publication-ready verification paragraph distinguishing these outcomes from historical Lean evidence, repository integration, and learner findings.
24. **New four-cycle.** Use ordered edges pq, qr, rs, sp, claims $(1/2, -1/3, 3/4, 1/6)$, and heights $(0, 1/2, 1/6, 11/12)$. Compute df, the residual, and its cycle sum. Choose a replacement closing claim that makes the table exact. State what an independently scored prediction on this unfamiliar carrier would contribute to a proposed transfer assessment, and what one such response cannot establish.

Scoring and remediation

The key gives four explicit points for each problem. Total and level scores locate work to revisit; this rubric is a proposed teaching aid, not a validated certification standard. Treat a wrong coefficient system, silent change of carrier or claims, unrecorded first response, or promotion of a replay into learner evidence as an error in that problem's scope point. Do not erase a correct calculation elsewhere.

Use the guide to repair the corresponding skill, then try a changed instance and retain both attempts. No grade grants authority to publish a learning-effect claim.